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98

Part A: Hash Table Definitions (Conceptual Understanding)

Q1. Define "collision" in the context of hash tables.

A1: 當不同的 key mapping 到同個 index 時就會發生的事

Q2. What is a "bucket" in a hash table?

A2: bucket 是 key mapping 到的地方，儲存資料。

Q3. Define "load factor (α)" and explain why it affects performance.

A3: $\alpha = \frac{\text{size}}{\text{size}}$ ，當 α 高時因為空間較少而造成 collision 機會變多，產生效能上的下降。
代表空間被使用的比率

Q4. What is "primary clustering," and which probing method suffers from it?

A4: 是指當連續長的一塊空間被占用時，造成需要多次 probing 的情形。
linear probing

Q5. What is "secondary clustering," and how is it different from primary clustering?

A5: 是指當 key 指到的位置時需要進行多次重覆的 probing 情形嚴重時。
被占用

兩者差異在一邊連續區塊的占用，另一個則是零錯7的占用。

Q6. Briefly explain the difference between:

- Open addressing
- Separate chaining

A6: 1. 在 collision 時用 probing 找到空位存值，使用一個大的 array 儲存。
array 上的

2. 在 collision 時會將值直接串聯在那個位置。

Part B: Hash Function Calculation (Collision & Pattern Observation)

Show your steps clearly.

Hash Function 1 — Division Method

$$h_1(k) = k \bmod 10$$

Hash Function 2 — Folding Method

Split key into two-digit chunks and sum the chunks.

$$h_2(k) = (\text{sum of 2-digit groups}) \bmod 11$$

Example:

Key = 8429 → groups: 84 + 29 → 113 → $113 \bmod 11 = 3$

Q7. (Compute using Hash Function 1)

Given keys: 27, 37, 47, 57, 67

Compute their hash values using:

$$h_1(k) = k \bmod 10$$

A7: 7, 7, 7, 7, 7

Q8. (Identify collision pattern)

From your results in Q7:

- What pattern do you observe?
- Explain why these keys collide.

A8: 1. 每個值的結果都一樣

2. 因為這些數字對 10 mod 之後 都剩 7, 且 10 是非質數

Q9. (Compute using Hash Function 2)

Compute $h_2(k)$ for: 1234, 9217, 4519, 9902

A9:

$$12+34=46, 46 \bmod 11 = 2$$

$$92+17=109, 109 \bmod 11 = 10$$

$$45+19=64, 64 \bmod 11 = 9$$

$$99+02=101, 101 \bmod 11 = 2$$

A: 2, 10, 9, 2

Q10. (Compare distribution)

- Which hash function (h_1 or h_2) produced more collisions for the input set?
- Which seems to spread keys more evenly?
- Provide 1–2 sentences of explanation.

A10: 1. h_1

2. h_2

3. 可以看到 h_1 的結果較為重複, h_2 則較為分散, 可能因為 h_1 用非質數 mod, h_2 用質數